

REPORT

Title

Prolog Program to Implement Best First Search Algorithm

Aim

To implement the **Best First Search algorithm in Prolog** using heuristic values to find a path from a start node to a goal node.

Problem Statement

Implement a Prolog program that performs Best First Search on a graph. The program should select nodes according to their heuristic values and find a path from the given starting node to the specified goal node.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp11.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp11.pl`.
3. Define the graph using `edge/2` facts.
4. Define heuristic values using `h/2`.
5. Implement Best First Search using a priority list.
6. Save and load the program using `[exp11]..`
7. Execute the query with the start and goal nodes.
8. Observe the generated path.

Program

```
% Graph
edge(a, b) .
edge(a, c) .
```

```

edge(b, d).
edge(b, e).
edge(c, f).
edge(c, g).
edge(d, h).
edge(e, h).
edge(f, g).
edge(g, h).

% Heuristic values
h(a, 7).
h(b, 6).
h(c, 5).
h(d, 4).
h(e, 3).
h(f, 4).
h(g, 2).
h(h, 0).

% Best First Search
best_first(Start, Goal, Path) :-
    search([node(Start, [Start])], Goal, Path).

search([node(Goal, Path)|_], Goal, Path).

search([node(Current, Path)|Rest], Goal, Solution) :-
    findall(
        node(Next, [Next|Path]),
        (edge(Current, Next),
         \+ member(Next, Path)),
        Children
    ),
    append(Rest, Children, NewList),
    sort_nodes(NewList, SortedList),
    search(SortedList, Goal, Solution).

% Sort nodes according to heuristic value
sort_nodes(Nodes, Sorted) :-
    predsort(compare_nodes, Nodes, Sorted).

compare_nodes(<, node(A,_), node(B,_)) :-
    h(A, HA),
    h(B, HB),
    HA < HB.

```

```
compare_nodes(>, node(A,_), node(B,_)) :-  
    h(A, HA),  
    h(B, HB),  
    HA > HB.
```

```
compare_nodes(=, _, _).
```

Sample Output

Query:

```
?- [exp11].  
true.
```

```
?- best_first(a, h, Path).
```

Output:

```
Path = [h, g, c, a].
```

Since the path is constructed from the goal back to the start, the actual path is:

```
a -> c -> g -> h
```

Conclusion

Thus, the **Best First Search algorithm** was successfully implemented in Prolog. The program uses heuristic values to select the most promising node and finds a path from the starting node to the goal node.